

DUVAN, Russian Federation

WMO: 285370

Lat: 55.716N

Lon: 57.899E

Elev: 338

StdP: 97.33

Time Zone: 5.00 (E05)

Period: 94-05

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.6	(c) -26.8	(d) -32.1	(e) 0.2	(f) -29.4	(g) -29.3	(h) 0.3	(i) -26.5	(j) 13.3	(k) -2.0	(l) 11.9	(m) -4.7	(n) 1.7	(o) 200	(p) 0.614

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	28.7	19.2	26.9	18.2	25.1	17.4	20.4	26.8	19.4	25.6	18.3	23.6	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.9	13.4	23.7	17.0	12.7	22.4	16.2	12.0	21.1	60.0	27.1	56.8	25.6	53.1	23.8	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 11.7	(b) 9.7	(c) 7.4	(e) -33.9	(f) 31.2	(g) 3.3	(h) 1.2	(i) -36.2	(j) 32.1	(k) -38.1	(l) 32.8	(m) -40.0	(n) 33.5	(o) -42.3	(p) 34.4		
(a) 11.7	(b) 9.7	(c) 7.4	(e) -34.0	(f) 21.7	(g) 3.3	(h) 1.6	(i) -36.3	(j) 22.8	(k) -38.2	(l) 23.8	(m) -40.1	(n) 24.6	(o) -42.4	(p) 25.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-12.4	-11.7	-5.2	3.1	10.8	15.8	17.9	15.2	9.5	3.5	-6.9	-12.2
	DBStd	12.44	6.78	7.36	4.82	5.78	5.10	4.17	3.67	4.08	4.56	4.77	6.93	7.61
	HDD10.0	3521	695	609	471	217	54	5	0	3	63	211	506	688
	HDD18.3	5918	953	842	729	457	237	98	54	118	268	461	756	946
	CDD10.0	731	0	0	0	11	79	178	246	163	47	8	0	0
	CDD18.3	87	0	0	0	0	3	21	42	19	2	0	0	0
	CDH23.3	812	0	0	0	2	58	208	368	158	18	0	0	0
	CDH26.7	179	0	0	0	0	3	48	79	44	5	0	0	0
Wind	WSAvg	2.6	2.6	2.9	2.9	2.8	3.2	2.6	2.1	2.0	2.3	2.8	2.5	2.9
Precipitation	PrecAvg	548	27	22	26	29	50	66	71	70	51	56	40	33
	PrecMax	756	58	50	58	62	112	190	183	134	106	109	92	91
	PrecMin	364	1	2	1	2	6	15	10	19	10	7	4	9
	PrecStd	114	16	12	15	16	27	39	39	29	25	29	22	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	1.7	6.5	22.2	26.9	30.5	30.7	30.4	25.6	19.4	5.6	0.9
		MCWB	-0.4	0.2	3.1	11.1	13.8	19.4	20.2	20.1	15.3	11.8	3.6	0.0
	2%	DB	-0.6	0.3	3.8	18.4	25.1	27.9	28.7	27.6	22.0	15.3	3.5	0.0
		MCWB	-1.4	-0.7	1.3	9.5	14.1	18.2	19.6	19.5	13.8	10.2	2.4	-0.5
	5%	DB	-2.0	-0.9	2.3	15.6	22.8	25.6	27.2	24.5	19.1	12.6	2.1	-0.9
		MCWB	-2.7	-1.7	0.3	8.8	13.7	17.5	19.3	17.8	13.0	9.0	0.9	-1.5
	10%	DB	-3.2	-2.0	1.1	12.0	20.2	23.5	25.4	21.7	16.5	10.2	1.1	-2.2
		MCWB	-3.8	-2.8	-0.4	6.9	12.3	16.6	18.4	16.5	11.8	7.7	0.3	-2.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.0	0.2	3.9	12.0	16.6	20.8	21.6	21.3	17.3	12.5	4.1	0.4
		MCDB	0.4	1.2	5.6	20.5	23.1	27.2	28.1	28.4	25.1	16.4	4.6	0.9
	2%	WB	-1.4	-0.5	1.7	10.4	15.3	19.6	20.7	19.7	14.8	11.2	2.5	-0.4
		MCDB	-0.8	0.2	3.2	17.1	22.2	26.3	27.0	26.4	19.6	15.0	3.2	-0.1
	5%	WB	-2.6	-1.6	0.7	8.9	14.1	18.2	19.8	18.3	13.4	9.5	1.4	-1.4
		MCDB	-2.0	-1.0	2.0	15.2	21.2	24.0	25.8	23.6	17.6	11.9	2.0	-1.0
	10%	WB	-3.9	-2.8	-0.2	7.2	13.0	17.1	18.9	17.0	12.2	8.0	0.4	-2.8
		MCDB	-3.2	-2.0	1.0	12.2	19.0	22.3	24.4	21.0	16.0	10.1	1.0	-2.3
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.9	7.4	9.1	10.2	10.1	10.1	8.5	8.1	5.4	5.4	6.0
		MCDBR	5.2	5.4	7.3	13.9	14.7	13.7	12.9	12.5	12.2	10.0	6.0	6.6
		MCWBR	5.0	4.8	5.2	7.1	6.6	6.1	5.5	5.4	5.8	5.8	5.1	6.2
	5% WB	MCDBR	5.5	4.9	6.0	13.6	13.4	12.4	11.6	10.9	10.7	9.3	4.9	6.5
		MCWBR	5.3	4.7	4.6	7.4	6.7	6.2	5.5	5.2	5.8	6.1	4.8	6.3
Clear-Sky Solar Irradiance	taub	0.290	0.314	0.369	0.407	0.401	0.393	0.415	0.432	0.380	0.351	0.312	0.288	
	taud	2.037	2.031	1.942	2.078	2.241	2.330	2.270	2.194	2.303	2.262	2.140	2.001	
	Ebn at Noon	581	723	766	804	839	851	821	774	768	687	556	471	
	Edh at Noon	66	98	138	140	127	118	123	125	97	79	60	54	
All-Sky Solar Radiation	RadAvg	0.72	1.69	3.06	4.23	5.11	5.52	5.45	4.23	2.68	1.21	0.58	0.43	
	RadStd	0.10	0.21	0.30	0.49	0.52	0.60	0.54	0.47	0.24	0.26	0.12	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page